

Algorithmic Trading and Quantitative Strategies

Part 8

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- 1 **Implementation of Factor Investing**
- 2 **Factor Investing in R**
- 3 **Factor Investing for Turkish Equities**
- 4 **The Secret To Successful Factor Investing**
- 5 **Database**

Implementation of Factor Investing

Factor Construction

Steps

- Factor ideas

 - Economic theory

 - Market insight

 - Market anomaly

 - Previous academic/industry research (see appendices)

- Data collection and cleaning

- Construction of factor variables

- Construction of factor mimicking portfolios

- Evaluating factor performance

- Backtesting factor based trading strategies

Factor Construction

Raw Data

- Stationarity

- Outliers

 - Trimming: the extreme values are discarded

 - Winsorization: the extreme values are instead replaced by certain percentiles

- Free from look ahead bias

- Free from survivorship bias

Transformations

Factors are often transformed such that the resulting series is symmetric or close to being normally distributed.

Frequently used transformations include natural logarithms, exponentials, and square roots.

For example, a factor such as market capitalization has a large skew because a sample of large cap stocks typically includes mega capitalization stocks. To reduce the influence of mega-capitalization companies, we may instead use the natural logarithm or square root of market capitalization.

Factor Construction

Standardization

Standardization rescales a variable while preserving its order

$$\hat{x}_i = \frac{x_i - \bar{x}}{\sigma_x}$$

where $\bar{x}$ and σ_x are *cross sectional* mean and standard deviation

Standardization allows better comparison across stocks since means and standard deviations are the same.

Standardization can be useful in combining multiple variables.

Factor Construction

Orthogonalization

Sometimes the performance of our factor might be related to another factor. Orthogonalizing a factor for other specified factor(s) removes this relationship. We can orthogonalize by using a) averages, b) running regressions or c) multidimensional portfolio sorts.

Demeaning using averages:

Typically used to calculate sector or country neutral factors

$\hat{x}_i = x_i - \mu$, where μ is the average factor value for the sector or country

Regression residuals:

$x_i = \alpha + \beta f_i + \epsilon_i$, where x_i is the original factor, f_i is the factor to orthogonalize x .

OLS ensures that ϵ is orthogonal to x

Our new orthogonalized factor is $\hat{x}_i = \epsilon_i$

Descriptive Analysis of Factor Data

Histograms

Descriptive statistics

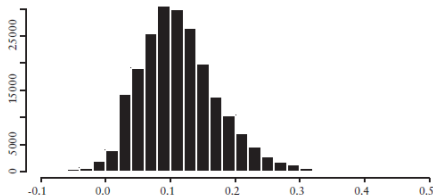
Outliers

Time series averages and dispersion

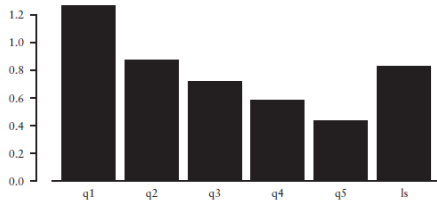
Descriptive Analysis of Factor Data

EXHIBIT 7.1 Portfolio Sorts Based on the EBITDA/EV Factor

Panel A: All Factor Values



Panel B: Monthly Average Returns for the Sorted Portfolios



Constructing Factor Mimicking Portfolios (FMP)

Types of Factor Models

Generic factor model:

$r_t^i = \alpha^i + \beta^{i,1} f_t^1 \dots + \beta^{i,K} f_t^K + \epsilon_t^i$, where β 's are factor exposures and f_t 's are factor returns

FMP are investable versions of the (potentially unobserved) factors. The goal of analyzing factor mimicking portfolios is to determine whether a factor earns a systematic premium.

Thus, three ingredients of factor investing are:

- Factor returns

- Factor exposures

- Factor mimicking portfolios

Constructing Factor Mimicking Portfolios (FMP)

Types of Factor Models

Fundamental Factor Models:

Based on (cross-Sectional) firm characteristics, such as market cap, P/E, etc.

Two methods can be used:

Portfolio Approach: Factors are constructed using sorted portfolios based on firm characteristics. Thus factor returns are observed. FMP's are the same as factors. Factor exposures are estimated using time series regressions.

Cross Sectional Regression Approach: Factor exposures are calculated from firm characteristics. Factor returns and FMP's are estimated using cross sectional regressions.

Macroeconomic Factor Models:

Macroeconomic variables and/or market indices are used as factors. Thus factor returns are observed. Factor exposures are always estimated using time series regressions. Three methods can be used:

Portfolio Approach: FMP's are constructed using sorted portfolios.

Cross Sectional Regression Approach: FMP's are estimated using cross sectional regressions.

Time Series Regression Approach: FMP's are found by using time series regressions where factor returns are directly regressed on asset returns.

Constructing Factor Mimicking Portfolios (FMP)

Types of Factor Models

Statistical Factor Models:

Purely based on latent variable models (e.g. principal component analysis, exploratory factor analysis) in statistics.

Neither factor returns, nor factor exposures are observed. Factors (same as FMP's) are estimated using principal component analysis. Factor exposures are estimated using time series regressions.

Constructing Factor Mimicking Portfolios (FMP)

Portfolio Approach

Sort the assets according to the selected factor (lowest to highest / highest to lowest). Group the sorted assets into N portfolios (usually $N = 5$, or $N = 10$). Example: 10 buckets according to market capitalization into 10 buckets. Each bucket contains stocks with similar market capitalization.

Construct portfolios from each bucket. Potential weighting schemes include:

Equal weighted: $w_i \propto +1$ if in long portfolio, $w_i \propto -1$ if in short portfolio

Market capitalization weighted: $w_i \propto +MCap$ if in long portfolio, $w_i \propto -MCap$ if in short portfolio

Score weighted (e.g. z-score): $w_i \propto x_i$

Rank weighted: $w_i \propto rank(x_i) - \bar{rank}(x)$

Distance weighted: $w_i \propto 1 - \frac{\max(x) - x_i}{\max(x) - \min(x)}$ if in long portfolio,

$w_i \propto \frac{x_i - \min(x)}{\max(x) - \min(x)} - 1$ if in short portfolio

The portfolios provide a representation of how returns vary across the different values of a factor. A FMP can be constructed by taking long position in the top bucket portfolio and taking short position in the bottom bucket portfolio. It is a zero-cost long-short portfolio.

Constructing Factor Mimicking Portfolios (FMP)

Portfolio Sorts

The portfolio sort methodology has several advantages.

The approach is easy to implement and can easily handle stocks that drop out or enter into the sample.

The resulting portfolios diversify away idiosyncratic risk of individual assets and provide a way of assessing how average returns differ across different magnitudes of a factor.

The portfolio sort methodology has several disadvantages.

The resulting portfolios may be exposed to **other risks factors** beyond the factor the portfolio was sorted on. In those instances, it is difficult to know which risk characteristics have an impact on the portfolio returns.

Because portfolio sorts are nonparametric, they do not give insight as to the functional form of the relation between the average portfolio returns and the factor.

Constructing Factor Mimicking Portfolios (FMP)

Cross Sectional Regression Approach

Regress stock returns on firm characteristics (e.g. z-score of dividend yield):

$$r_{t:t+1}^i = \alpha^i + \beta_1^i f_t^1 + \dots + \beta_N^i f_t^N + \epsilon_t^i$$

Here β 's are (**observed**) firm characteristics, f are **pure** factor returns (that need to be **estimated**) corresponding to firm characteristics.

In matrix form, the cross sectional regression can be written as:

$$R = \beta^T F + \epsilon$$

Constructing Factor Mimicking Portfolios (FMP)

Cross Sectional Regression Approach

Since error variances may differ for different firms, there is heteroskedasticity problem. Thus we use generalized least squares (GLS), rather than ordinary least squares (OLS). We assume zero correlation among residuals.

For the regression $R = \beta^T F + \epsilon$, we have:

OLS: Assumes $\text{cov}(\epsilon) = \sigma_\epsilon^2 \cdot \mathbb{I}$. Estimate is given by $\hat{F} = (\beta^T \beta)^{-1} \beta^T R$

GLS: Assumes $\text{cov}(\epsilon) = D = \text{diag}(\sigma_{\epsilon_1}^2, \dots, \sigma_{\epsilon_N}^2)$. Estimate is given by $\hat{F} = (\beta^T \hat{D}^{-1} \beta)^{-1} \beta^T \hat{D}^{-1} R$, where $\hat{D} = \text{diag}(\hat{\sigma}_{\epsilon_1}^2, \dots, \hat{\sigma}_{\epsilon_N}^2)$

The equation is typically estimated with weighted least squares (WLS) using inverse specific volatility, market caps or squared market caps as weights.

Constructing Factor Mimicking Portfolios (FMP)

Cross Sectional Regression Approach

The estimate of F here has an interesting interpretation as the return on a portfolio $h = (h_1, \dots, h_N)$ that solves:

$$\min_h \frac{1}{2} h^T D h$$

subject to $h^T \beta = 1$ and $D = \text{diag}(\hat{\sigma}_{\epsilon_1}^2, \dots, \hat{\sigma}_{\epsilon_N}^2)$

The portfolio h minimizes asset return residual variance subject to having unit exposure to the attribute F

The estimated factor realization is then the portfolio return:

$$\hat{F} = h^T R$$

Thus, h represents the factor mimicking portfolio. To find the weights by normalizing h such that $\sum_{i=1}^N h_i = 1$, i.e. $FMC = \left\{ \frac{h_i}{\sum_{i=1}^N h_i} \right\}$

Constructing Factor Mimicking Portfolios (FMP)

Time Series Regression Approach

This approach can only be used when the factors are observed, i.e. only for macroeconomic factor models.

FMP's are found by using time series regressions where factor returns are directly regressed on asset returns:

$$f_t = \gamma_0 + \sum_{i=1}^N \gamma_i r_t^i + \eta_t$$

Typically not all assets are used in time series regressions. Rather, first portfolios that have possibly distinct characteristics are formed. This can be done using the sorting technique in portfolio approach. Then these **base assets** are used in regressions.

Estimated regression loadings are used as the weights in FMP's:

$$FMP = \left\{ \frac{\hat{\gamma}}{\sum_{i=1}^N \hat{\gamma}_i} \right\}$$

Constructing Factor Mimicking Portfolios (FMP)

Statistical Factor Approach

Principal component analysis (PCA) is a statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called principal components.

The number of principal components is less than or equal to the number of original variables.

This transformation is defined in such a way that the first principal component has the largest possible variance (that is, accounts for as much of the variability in the data as possible), and each succeeding component in turn has the highest variance possible under the constraint that it is orthogonal to the preceding components.

The resulting vectors are an uncorrelated orthogonal basis set.

Constructing Factor Mimicking Portfolios (FMP)

Statistical Factor Approach

Instead of theorizing on risk factors, this type of factor model tries to determine the risk factors by observing them in the data.

It has the advantages of being easy to understand, compute and visualize, but on the other hand, it is more sensitive to the data used to compute the model and factors don't have real life correspondance.

The only parameter of this model is the number of factors to include. The number of factors is usually determined by running a principal component analysis on the available data and observing share of variation explained by each factor. -To perform the principal component analysis, one calculates the eigenvectors V and eigenvalues λ of the sample data matrix by decomposing the covarinace matrix:

$$Cov(R) = V\lambda V^T$$

Multifactor Models



↓
Value
Factor



↓
Momentum
Factor



↓
Size
Factor



↓
Growth
Factor



↓
Profitability
Factor



↓
Sentiment
Factor



Composite
Quant Factor

Multifactor Models



Evaluating Factor Premiums

Information Coefficient

The IC is a linear statistic that measures the cross-sectional correlation between a factor and its subsequent realized return: $IC_{t:t+1} = \text{corr}(f_t, r_{t:t+1})$, where f_t is a vector of cross sectional factor values at time t and $r_{t:t+1}$ is a vector of returns over the time period t to $t + 1$.

We can also use rank correlation

This reduces the undue influence of outliers and reduces the influence of variables with unequal variances

$$IC_{t:t+1} = \text{corr}(\text{rank}(f_t), r_{t:t+1})$$

$$IC_{t:t+1} = \text{corr}(\text{rank}(f_t), \text{rank}(r_{t:t+1}))$$

The values of the IC range from -1 to +1. A positive IC indicates a positive relation between the factor and return. A negative IC indicates a negative relation between the factor and return.

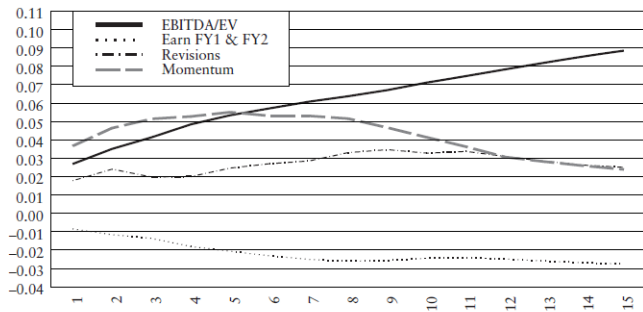
ICs are usually calculated over an interval, for example, daily or monthly. We can evaluate how a factor has performed by examining the time series behavior of the ICs.

Looking at the mean IC tells how predictive the factor has been over time.

Evaluating Factor Premiums

Information Coefficient

EXHIBIT 7.4 Information Coefficients over Various Horizons for EBITDA/EV, Growth of Fiscal Year 1 and Fiscal Year 2 Earnings Estimates, Revisions, and Momentum Factors



Evaluating Factor Premiums

Fama-MacBeth Approach

Using cross sectional regression approach estimate factor returns

To evaluate these to determine the significance of the cross-sectional regression, calculate the mean and standard errors of the time series of factor returns:

$$\hat{f} = \frac{1}{T} \sum_{t=1}^T \hat{f}_t, \hat{\sigma}_f^2 = \frac{1}{T^2} \sum_{t=1}^T (\hat{f}_t - \hat{f})^2$$

Perform a t-test on factor returns to test whether they are statistically significantly different from zero or not: $t = \hat{f} / (\sqrt{N \cdot \hat{\sigma}_f^2})$

Evaluating Factor Premiums

Portfolio Analysis

After constructing FMP's, analyze their performance to assess the factor premiums

In portfolio approach we can analyze each quintile/decile portfolio as well

Two types of comparison:

- Compare the long-only factor portfolio with a passive benchmark

- Compare the long-short factor portfolio with a cash benchmark

Common performance metrics:

- Annualized Return

- Annualized Volatility

- Sharpe or Information Ratio

- Maximum Drawdown

- Hit Ratio

Evaluating Factor Premiums

Portfolio Analysis

By studying the return behavior of the factor portfolios, we may assess the return and risk profile of the factor.

In some cases, we may identify a monotonic relationship of the returns across the portfolios.

In other cases, we may identify a large difference in returns between the extreme portfolios.

Still in other cases, there may be no relationship between the portfolio returns.

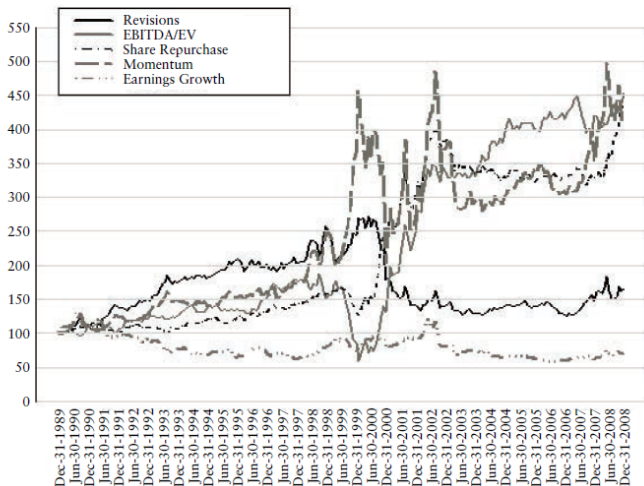
Overall, the return behavior of the portfolios will help us conclude whether there is a premium associated with a factor and describe its properties.

The standard statistical testing procedure for portfolios sorts is to use a Student's t-test to evaluate the significance of the mean return differential between the portfolios of stocks with the highest and lowest values of the factor.

Evaluating Factor Premiums

Factor Mimicking Portfolio Analysis

EXHIBIT 7.6 Cumulative Returns of Long-Short Portfolios



Commercial Products for Factor Research

S&P Capital IQ

Factset

Compustat

Reuters Starmine

Sell Side Research Products

see JP Morgan Factor Handbook

Evaluating Factor Premiums

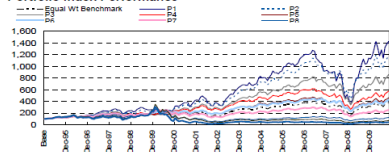
Composite Value in Bottom 2000 Universe

Rebalance every 1 month(s)

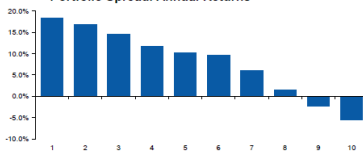
5 Year(s): 11/30/1995 to 11/30/2000 Portfolio Statistics					5 Year(s): 11/30/2000 to 11/30/2005 Portfolio Statistics					5 Year(s): 11/30/2005 to 11/30/2010 Portfolio Statistics					Total Period: 1/31/1995 to 11/30/2010 Portfolio Statistics				
Port	Avg Ret	Ann Ret	Std Dev	% Out Perf.	Port	Avg Ret	Ann Ret	Std Dev	% Out Perf.	Port	Avg Ret	Ann Ret	Std Dev	% Out Perf.	Port	Avg Ret	Ann Ret	Std Dev	% Out Perf.
1	1.3%	14.4%	6%	65%	1	2.6%	31.3%	6%	70%	1	1.5%	9.5%	12%	58%	1	1.8%	18.3%	9%	63%
2	1.3%	14.4%	5%	58%	2	2.3%	28.2%	6%	62%	2	1.0%	7.8%	9%	62%	2	1.5%	16.9%	7%	59%
3	1.0%	11.6%	5%	48%	3	2.0%	23.8%	6%	60%	3	0.9%	7.4%	8%	65%	3	1.3%	14.6%	6%	57%
4	1.1%	12.0%	5%	45%	4	1.6%	19.5%	5%	57%	4	0.5%	2.8%	7%	50%	4	1.1%	11.7%	6%	48%
5	0.8%	9.7%	5%	40%	5	1.5%	18.4%	5%	58%	5	0.5%	3.6%	7%	52%	5	1.0%	10.1%	5%	49%
6	0.9%	9.8%	6%	52%	6	1.2%	13.7%	6%	55%	6	0.4%	2.8%	6%	45%	6	0.9%	9.6%	6%	50%
7	0.8%	6.9%	7%	53%	7	0.8%	7.0%	7%	38%	7	0.3%	0.7%	6%	43%	7	0.7%	6.0%	7%	44%
8	0.5%	0.9%	10%	48%	8	0.1%	-4.5%	9%	43%	8	0.5%	3.2%	7%	52%	8	0.5%	1.4%	8%	48%
9	0.5%	-1.8%	12%	52%	9	-0.1%	-8.1%	11%	33%	9	0.0%	-3.4%	7%	38%	9	0.3%	-2.2%	10%	43%
10	-0.1%	-10.1%	12%	47%	10	0.6%	-10.0%	19%	42%	10	0.3%	-1.4%	10%	47%	10	0.4%	-5.4%	14%	46%
Total Test					Total Test					Total Test					Total Test				
	Avg Ret	Rank	Avg IC	Avg Assets		Avg Ret	Rank	Avg IC	Avg Assets		Avg Ret	Rank	Avg IC	Avg Assets		Avg Ret	Rank	Avg IC	Avg Assets
Universe	0.8%	5.1%	3.5%	1619	Universe	1.3%	9.0%	6.5%	1763	Universe	0.6%	2.9%	2.3%	1542	Universe	1.0%	5.3%	3.7%	1763
Long Short Strategy Statistics Portfolio 1 less Portfolio 10					Long Short Strategy Statistics Portfolio 1 less Portfolio 10					Long Short Strategy Statistics Portfolio 1 less Portfolio 10					Long Short Strategy Statistics Portfolio 1 less Portfolio 10				
	Avg Ret	Ann Ret	Std Devn	% Out Perf.		Avg Ret	Ann Ret	Std Devn	% Out Perf.		Avg Ret	Ann Ret	Std Devn	% Out Perf.		Avg Ret	Ann Ret	Std Devn	% Out Perf.
Long/Short	1.4%	11.8%	10%	53%	Long/Short	2.0%	10.2%	14%	60%	Long/Short	1.1%	12.1%	6%	63%	Long/Short	1.4%	10.1%	10.0%	58%
	T-Stat			Avg Assets		T-Stat			Avg Assets		T-Stat			Avg Assets		T-Stat			Avg Assets
Long/Short	1.12			325	Long/Short	1.12			353	Long/Short	1.47			389	Long/Short	1.90			353

Evaluating Factor Premiums

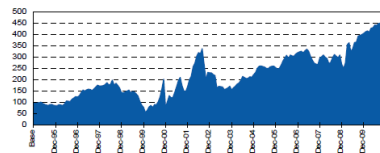
Portfolio Index Performance



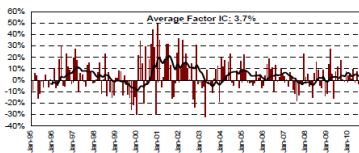
Portfolio Spread. Annual Returns



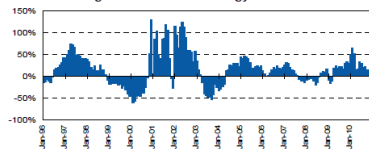
Cumulative Returns



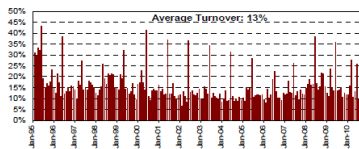
Information Co-Efficients (IC)



12 Month Rolling Returns Of L/S Strategy



Turnover within Portfolio 1



Factor Investing in R

Factor Data

```
suppressMessages(library(zoo))
```

```
# Load Data
```

```
load("D:/GOOGLE DRIVE/ACADEMIC/LECTURES/BOUN_2018_2/Lecture_Notes/Pa
```

```
# ROE Factor
```

```
head(names(TrailROE),5)
```

```
## [1] "XU100" "XU030" "XBANK" "XUSIN" "AKBNK"
```

```
Factor.Data<-TrailROE[,-c(1:4)] #Remove BIST indices
```

```
# Use monthly data
```

```
Month.Ends<-xts::endpoints(Factor.Data, on="months")
```

```
Month.Ends.Dates<-time(Factor.Data)[Month.Ends[-1]]
```

```
Factor.Data<-window(Factor.Data,Month.Ends.Dates)#Convert to monthly
```

```
head(Factor.Data,5)
```

```
##
```

```
AKBNK
```

```
GARAN
```

```
ISCTR
```

```
VAKBN
```

```
HALKB
```

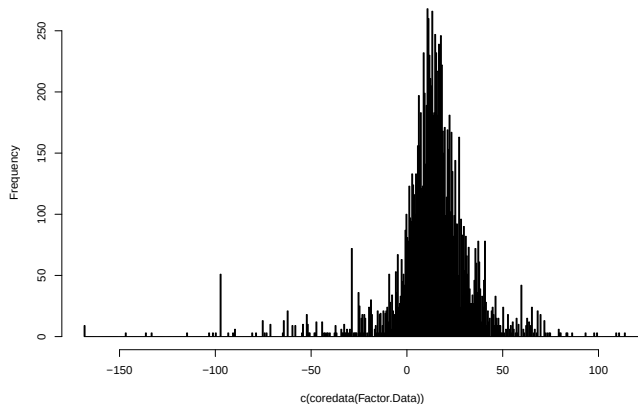
```
YKBNK
```

```
TSKB SK
```

Factor Data

```
# Plot data  
hist(c(coredata(Factor.Data)),1000)
```

Histogram of c(coredata(Factor.Data))



Factor Data

```
# Winsorize Outliers  
dat<-coredata(Factor.Data)#convert to matrix  
dat[dat< -100]<- -100  
dat[dat> 100]<- 100  
Factor.Data<-zoo(dat,time(Factor.Data))
```

Factor Data

```
# Standardize
```

```
Z.Score<-function(x) (x-mean(x,na.rm=T))/sd(x,na.rm=T)
```

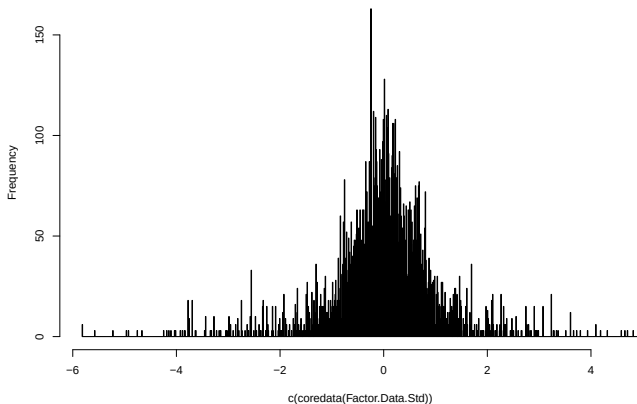
```
Factor.Data.Std<-t(apply(Factor.Data,1,Z.Score)) #gives matrix
```

```
Factor.Data.Std<-zoo(Factor.Data.Std,time(Factor.Data))
```

Factor Data

```
# Plot standardized data  
hist(c(coredata(Factor.Data.Std)),1000)
```

Histogram of c(coredata(Factor.Data.Std))



Information Coefficient

Get prices and returns

```
Prices<-window(na.locf(Close[, -c(1:4)]), Month.Ends.Dates)
Rets<-Prices/lag(Prices, -1, na.pad = T) - 1
```

```
Bench<-window(na.locf(Close[, "XU100"]), Month.Ends.Dates)
Bench.Rets<-Bench/lag(Bench, -1, na.pad = T) - 1
```

Calculate IC

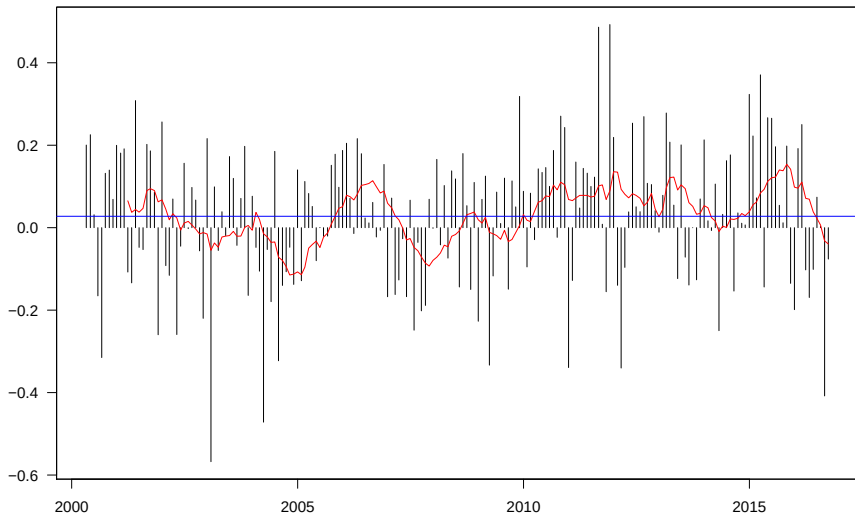
```
IC<-vapply(2:dim(Rets)[1],
           function(i) cor(as.numeric(Factor.Data.Std[i-1,]),
                           as.numeric(Rets[i,]),
                           use="pairwise.complete.obs"),
           3.54)
IC<-zoo(IC, time(Rets)[-1])
```

Information Coefficient

```
#Plot IC  
plot(IC,type="h",main="Info Coefficient",las=1,xlab="",ylab="")  
lines(rollmeanr(IC,12),col=2)  
abline(h=mean(IC),col=4)  
title(sub=paste("Avg IC = ",round(100*mean(IC),1),"%",sep=""))
```

Information Coefficient

Info Coefficient



Information Coefficient

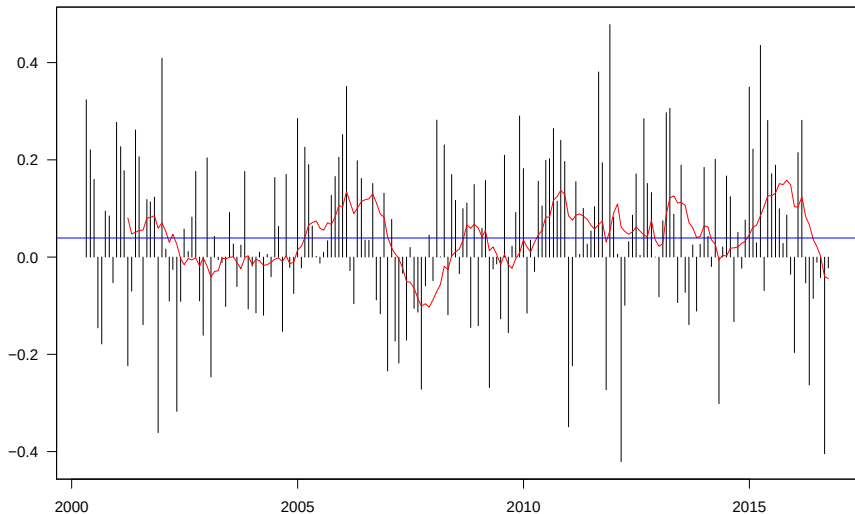
```
# Calculate Rank IC
IC.Rank<-vapply(2:dim(Rets)[1],
  function(i) cor(
    rank(as.numeric(Factor.Data.Std[i-1,]),na.last="keep"),
    as.numeric(Rets[i,]),
    use="pairwise.complete.obs"),
  3.54)
IC.Rank<-zoo(IC.Rank,time(Rets)[-1])
```

Information Coefficient

```
#Plot IC  
plot(IC.Rank,type="h",main="Rank Info Coefficient",  
      las=1,xlab="",ylab="")  
lines(rollmeanr(IC.Rank,12),col=2)  
abline(h=mean(IC.Rank),col=4)  
title(sub=paste("Avg Rank IC = ",round(100*mean(IC.Rank),1),  
                "%",sep=""))
```

Information Coefficient

Rank Info Coefficient



Factor Portfolios

```
# Construct 5 portfolios
```

```
suppressMessages(library(ggplot2))
```

```
Find.Portfolios<-function(x) as.numeric(cut_number(x, n = 5))
```

```
Port.IDs<-t(apply(Factor.Data.Std,1,Find.Portfolios))
```

```
Port.IDs<-zoo(Port.IDs,time(Factor.Data.Std))
```

```
head(Port.IDs,5) # Stock i belongs to which portfolio
```

```
##
```

```
## 2000-03-31 5 4 4 NA NA 4 4 NA NA 5 NA NA 2 1 1 2 5 4 NA NA 3 1 2
```

```
## 2000-04-28 5 4 4 NA NA 4 4 NA NA 5 NA NA 2 1 1 2 5 4 NA NA 3 1 2
```

```
## 2000-05-31 5 4 4 NA NA 4 4 NA NA 5 NA NA 2 1 1 2 5 4 NA NA 3 1 2
```

```
## 2000-06-30 5 4 4 NA NA 5 4 NA NA 4 NA NA 2 1 1 2 5 4 NA NA 1 1 3
```

```
## 2000-07-31 5 4 4 NA NA 5 4 NA NA 4 NA NA 2 1 1 2 5 4 NA NA 1 1 3
```

```
##
```

```
## 2000-03-31 5 NA 5 5 5 NA NA 4 4 NA 1 NA 1 NA NA NA NA NA NA NA
```

```
## 2000-04-28 5 NA 5 5 5 NA NA 4 4 NA 1 NA 1 NA NA NA NA NA NA NA
```

Factor Portfolios

```
# Fnc for portfolio return
Construct.Port<-function(ID){
  Weights<-(Port.IDs==ID) #Weights
  Norm.Weights<-t(apply(Weights,1,function(x) x/sum(x,na.rm=T)))
  Norm.Weights<-zoo(Norm.Weights,time(Weights))#Normalized

  Norm.Weights<-lag(Norm.Weights,-1)#Lag weights

#Calculate portfolio rets (no fees!!)
  Port.Rets<-apply(Norm.Weights*Rets[-1,],1,sum,na.rm=T)
  Port.Rets<-zoo(Port.Rets,time(Rets)[-1])
  Port<-cumprod(1+Port.Rets)
  Port}
```

Factor Portfolios

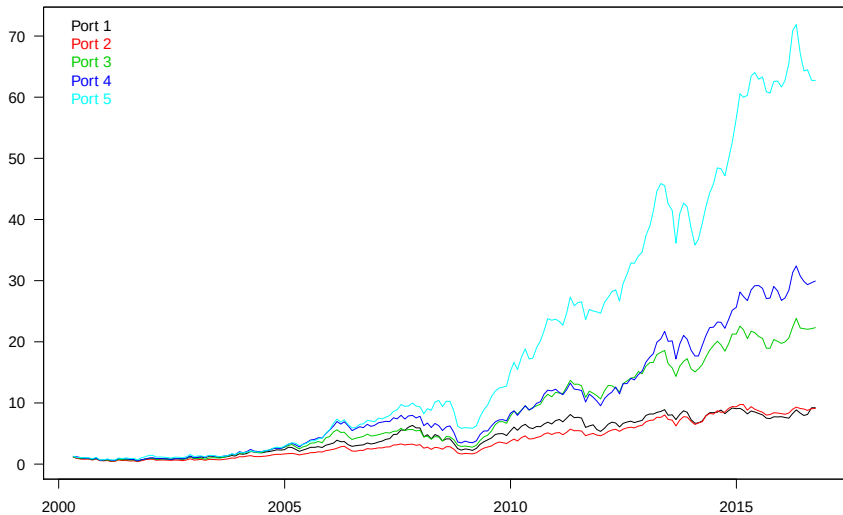
```
# Construct decile portfolios  
P1<-Construct.Port(1)  
P2<-Construct.Port(2)  
P3<-Construct.Port(3)  
P4<-Construct.Port(4)  
P5<-Construct.Port(5)  
  
All.Portfolios<-merge(P1,P2,P3,P4,P5)
```

Factor Portfolios

```
plot(All.Portfolios,screens=1,col=1:5,  
     xlab="",ylab="",las=1,main="Factor Portfolios")  
legend("topleft",paste("Port ",1:5,sep=""),text.col=1:5,bty="n")  
  
plot(All.Portfolios,screens=1,col=1:5,  
     xlab="",ylab="",las=1,log="y",main="Factor Portfolios (Log Scale)")  
legend("topleft",paste("Port ",1:5,sep=""),text.col=1:5,bty="n")
```

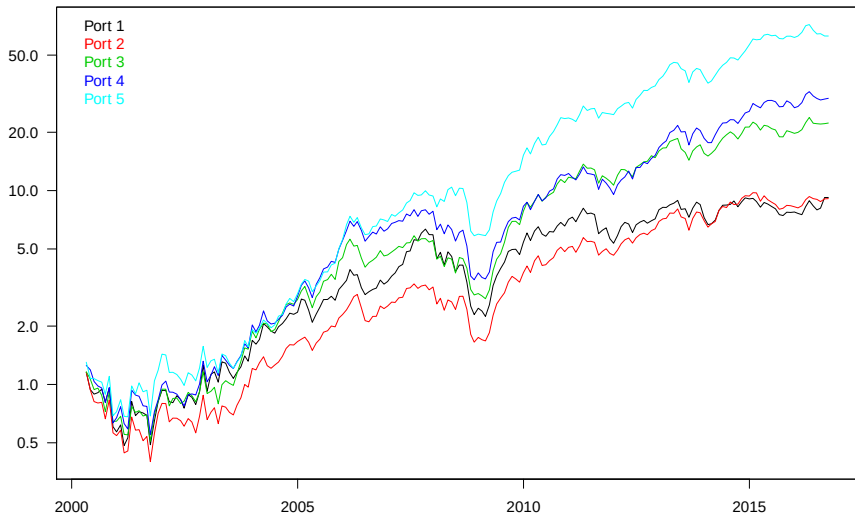
Factor Portfolios

Factor Portfolios



Factor Portfolios

Factor Portfolios (Log Scale)



Factor Portfolios

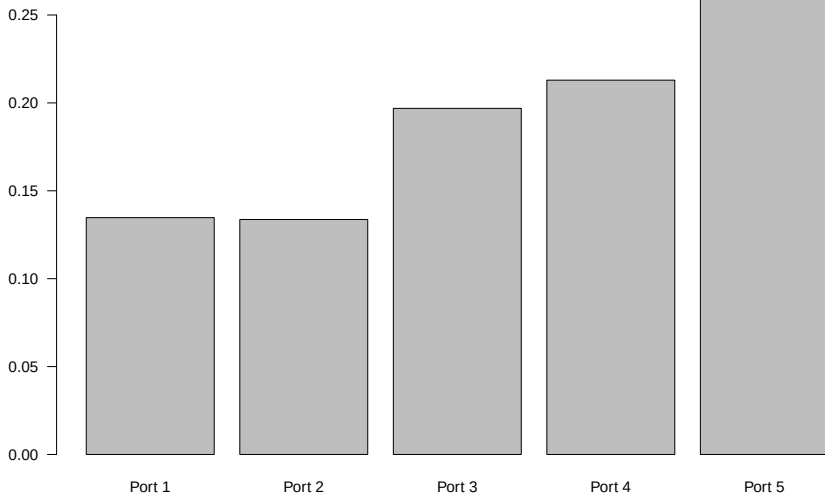
```
# Fnc for CAGR return
CAGR<-function(myzoo){
  (tail(myzoo,1)/as.numeric(head(myzoo,1)))^
    (365/as.numeric(diff(range(time(myzoo))))))-1
}
```

```
All.CAGR<-cbind(CAGR(All.Portfolios[,1]),
                CAGR(All.Portfolios[,2]),
                CAGR(All.Portfolios[,3]),
                CAGR(All.Portfolios[,4]),
                CAGR(All.Portfolios[,5]))
```

```
barplot(as.numeric(All.CAGR),names.arg=paste("Port ",1:5,sep=""),
        main="CAGR",las=1)
```

Factor Portfolios

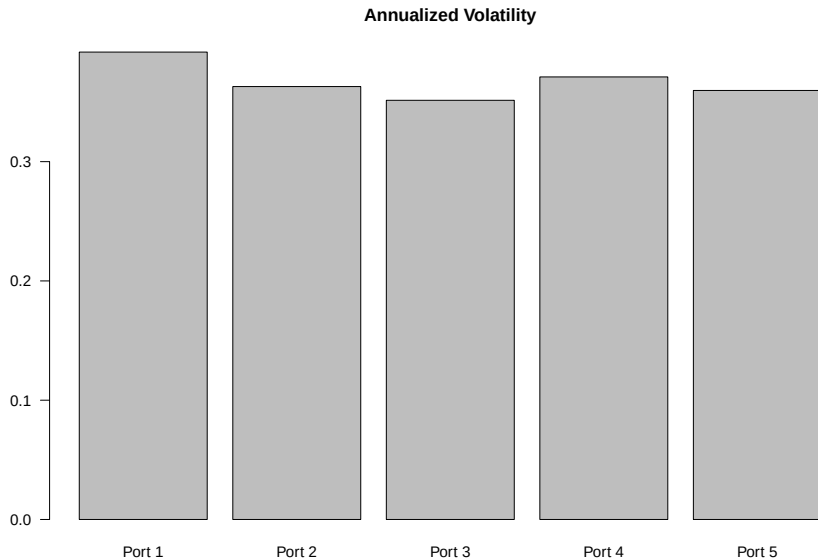
CAGR



Factor Portfolios

```
All.Vol<-apply(All.Portfolios,2,  
               function(x) sqrt(12)*sd(diff(log(x))))  
  
barplot(All.Vol,names.arg=paste("Port ",1:5,sep=""),  
        main="Annualized Volatility",las=1)
```

Factor Portfolios



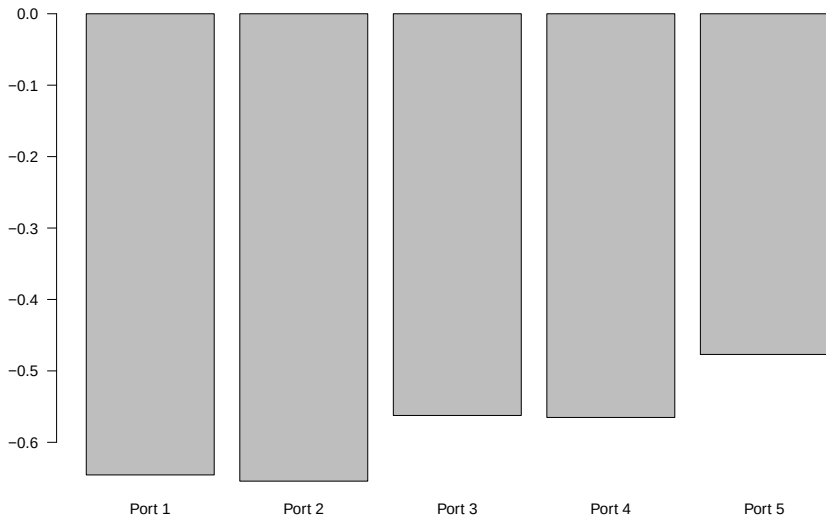
Factor Portfolios

```
# Fnc for max drawdown
MaxDD<-function(myzoo){
  min(myzoo/TTR::runMax(myzoo, n=1,cumulative =T)-1,na.rm=T)
}

All.MDD<-apply(All.Portfolios,2,MaxDD)
barplot(All.MDD,names.arg=paste("Port ",1:5,sep=""),
        main="Max Drawdowns",las=1)
```

Factor Portfolios

Max Drawdowns



FMP (Portfolio Approach)

```
# Construct FMP
```

```
FMP.Rets<-(P5/lag(P5,-1)-1)-(P1/lag(P1,-1)-1)
```

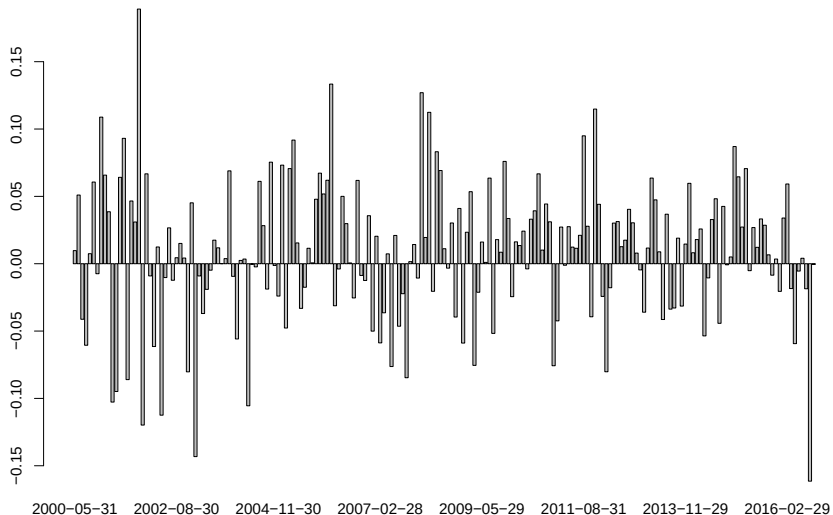
```
FMP<-cumprod(1+FMP.Rets)
```

```
barplot(FMP.Rets, main="FMP Returns")
```

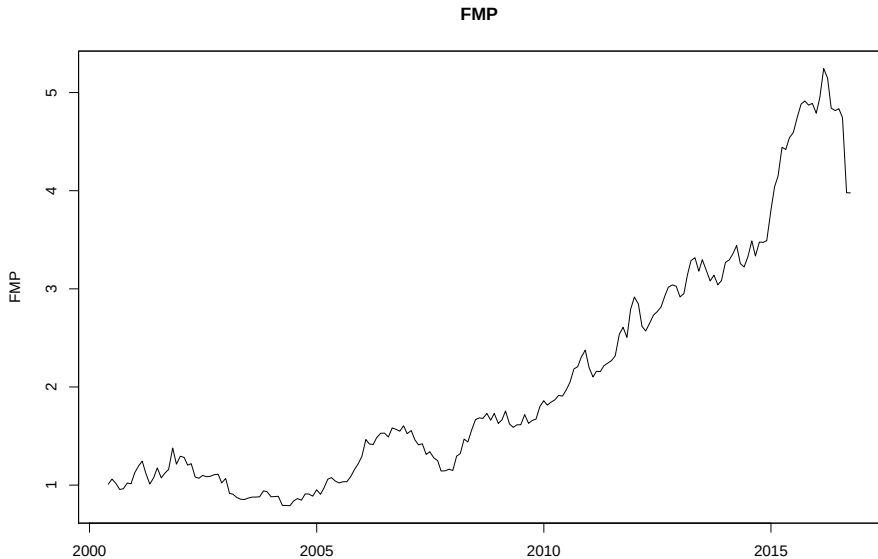
```
plot(FMP, main="FMP")
```

FMP (Portfolio Approach)

FMP Returns



FMP (Portfolio Approach)



FMP (Portfolio Approach)

```
# t-test
t.test(FMP.Rets, alternative="greater")

##
## One Sample t-test
##
## data: FMP.Rets
## t = 2.3085, df = 196, p-value = 0.01101
## alternative hypothesis: true mean is greater than 0
## 95 percent confidence interval:
##  0.002359265      Inf
## sample estimates:
##  mean of x
## 0.008304495
```

FMP (Cross Sectional Regression Approach)

```
# Fnc for obtaining regression slope
Get.Slope<-function(i){
  dat<-cbind(as.numeric(Rets[i,]),
             as.numeric(Factor.Data.Std[i-1,]))
  dat<-na.omit(dat)

  fit<-lm(dat[,1]~dat[,2])
  summary(fit)$coefficients[2,1]

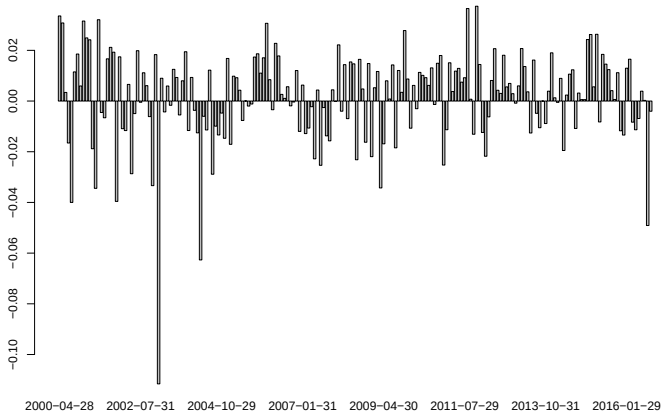
}

# Obtain factor returns through cross sectional regressions
FMP.CS.Rets<-vapply(2:dim(Rets)[1],Get.Slope,3.54)
FMP.CS.Rets<-zoo(FMP.CS.Rets,time(Rets)[-1])

# FMP
FMP.CS<-cumprod(1+FMP.CS.Rets)
```

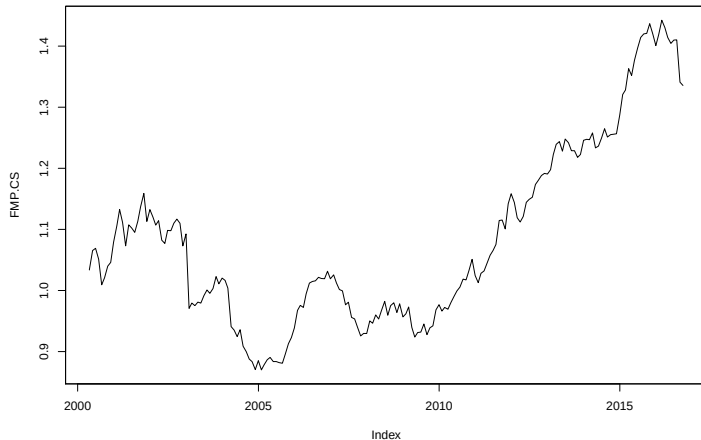
FMP (Cross Sectional Regression Approach)

```
barplot(FMP.CS.Rets)
```



FMP (Cross Sectional Regression Approach)

```
plot(FMP.CS)
```



FMP (Cross Sectional Regression Approach)

```
# t-test
t.test(FMP.CS.Rets, alternative="greater")

##
## One Sample t-test
##
## data: FMP.CS.Rets
## t = 1.28, df = 197, p-value = 0.101
## alternative hypothesis: true mean is greater than 0
## 95 percent confidence interval:
## -0.0004729014 Inf
## sample estimates:
## mean of x
## 0.001624493
```

FMP (Cross Sectional Regression Approach)

Fnc for obtaining FMP weights

```
Get.FMP.Weights<-function(i){
  dat<-cbind(as.numeric(Rets[i,]),
             as.numeric(Factor.Data.Std[i-1,]))
  non.nas<-which(apply(dat,1,function(x) (!is.na(x[1])) & (!is.na(x[2]))))
  dat<-na.omit(dat)
```

```
XX<-cbind(rep(1,dim(dat)[1]),dat[,2])
WW<-t(solve(t(XX)%*%XX)%*%t(XX))[,2]
```

```
W<-rep(0,dim(Rets)[2])
W[non.nas]<-WW
W
}
```

Obtain factor returns through cross sectional regressions

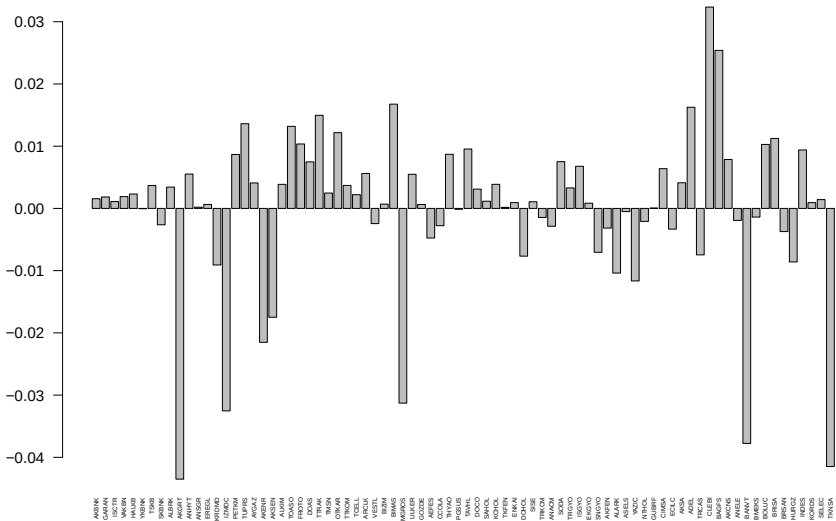
```
FMP.W<-t(vapply(2:dim(Rets)[1],Get.FMP.Weights,rep(3.54,dim(Rets)[2]))
FMP.W<-zoo(FMP.W,time(Rets)[-1])
```

FMP (Cross Sectional Regression Approach)

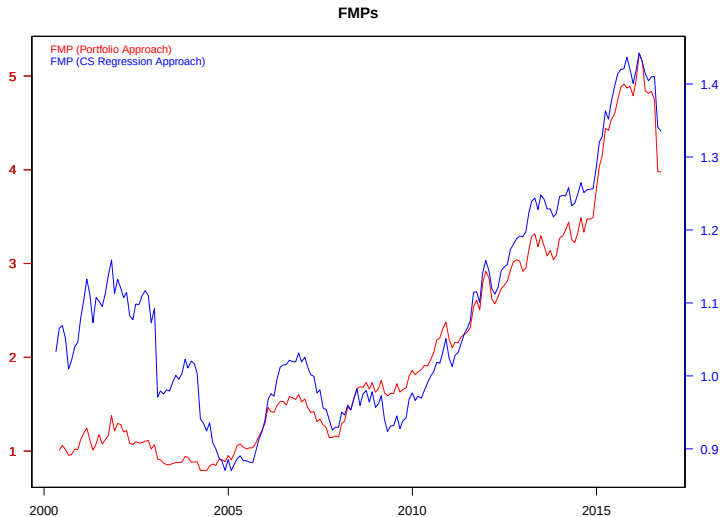
```
barplot(as.numeric(tail(FMP.W,1)),  
        main="FMP Weights for Last Period",las=2,  
        names.arg=names(Rets),cex.names=0.4)
```

FMP (Cross Sectional Regression Approach)

FMP Weights for Last Period



FMP Comparison



FMP Comparison

```
# Volatility and correlation of FMPs
```

```
sqrt(12)*sd(FMP.Rets)
```

```
## [1] 0.1749072
```

```
sqrt(12)*sd(FMP.CS.Rets)
```

```
## [1] 0.06186271
```

```
cor(FMP.Rets,FMP.CS.Rets[-1])
```

```
## [1] 0.8027009
```

Time Series Regression Approach

```

# Use USDTRY as macro factor
TRY<-window(FX[, "Turkey"], Month.Ends.Dates)
TRY.Rets<-TRY/lag(TRY, -1, na.pad=T)-1

# Use P1-P5 as base assets
Base.Asset.Rets<-All.Portfolios

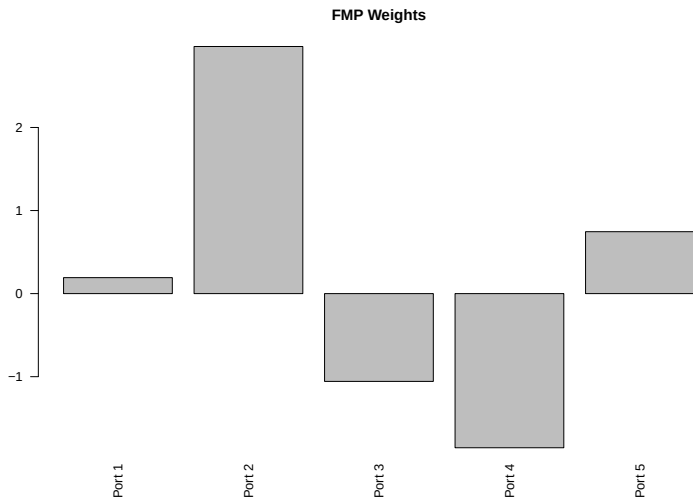
# Fnc to compute weights
Get.TS.Weights<-function(i){
  dat<-na.omit(merge(TRY.Rets[(i-35):i],
                    All.Portfolios[(i-36):(i-1),]))

  fit<-lm(dat[,1]~dat[, -1])
  W<-summary(fit)$coefficients[2:6,1]
  W/sum(W)}

FMP.W<-t(vapply(37:dim(Rets)[1], Get.TS.Weights, rep(3.54, 5)))
FMP.W<-zoo(FMP.W, time(Rets)[-c(1:36)])

```

Time Series Regression Approach



Statistical Factor Model

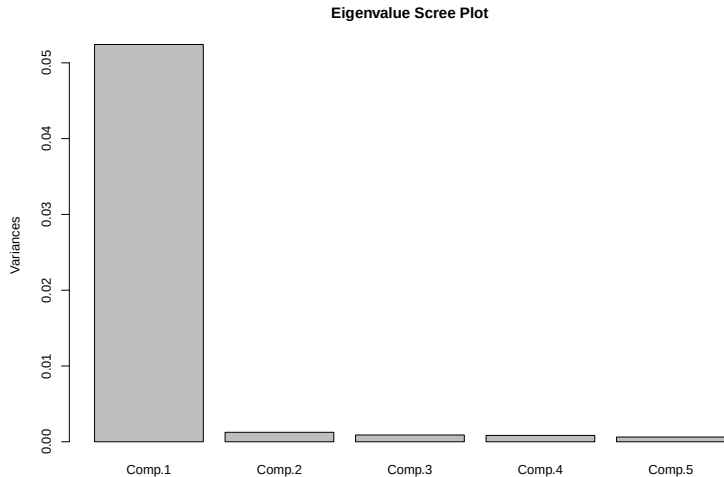
```
# Principal component analysis
```

```
PCA<-princomp(coredata(diff(log(All.Portfolios))))  
summary(PCA)
```

```
## Importance of components:
```

```
##               Comp.1      Comp.2      Comp.3      Comp.4  
## Standard deviation    0.2289558 0.03537128 0.02994258 0.02893382  
## Proportion of Variance 0.9355310 0.02232834 0.01600050 0.01494055  
## Cumulative Proportion 0.9355310 0.95785934 0.97385984 0.98880039  
##               Comp.5  
## Standard deviation    0.02505093  
## Proportion of Variance 0.01119961  
## Cumulative Proportion 1.00000000
```

Statistical Factor Model



Statistical Factor Model

```
loadings(PCA)
```

```
##
## Loadings:
##      Comp.1 Comp.2 Comp.3 Comp.4 Comp.5
## P1 -0.475  0.779 -0.252 -0.152 -0.283
## P2 -0.443  0.182  0.442  0.450  0.610
## P3 -0.427 -0.283  0.648 -0.272 -0.494
## P4 -0.453 -0.337 -0.344 -0.594  0.459
## P5 -0.436 -0.407 -0.450  0.590 -0.305
```

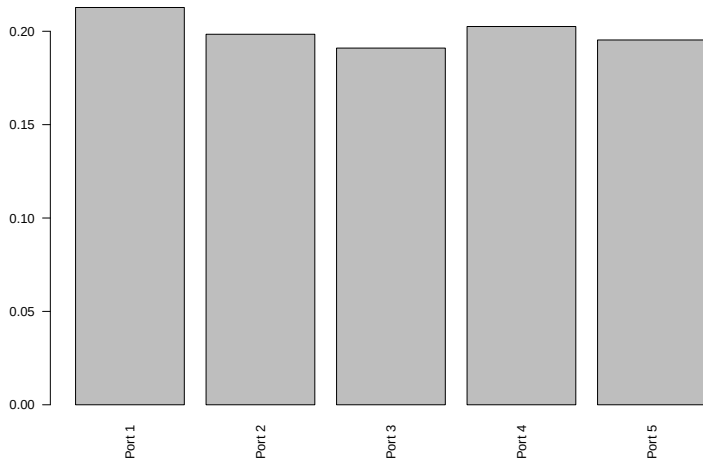
```
##
##              Comp.1 Comp.2 Comp.3 Comp.4 Comp.5
## SS loadings      1.0   1.0   1.0   1.0   1.0
## Proportion Var    0.2   0.2   0.2   0.2   0.2
## Cumulative Var    0.2   0.4   0.6   0.8   1.0
```

Statistical Factor Model

```
PCA.FMP<-loadings(PCA)[,1]  
PCA.FMP<-PCA.FMP/sum(PCA.FMP)
```

Statistical Factor Model

PCA FMP Weights



Statistical Factor Model

```
PCA.Factor.Rets<-zoo(PCA$scores,time(diff(log(All.Portfolios))))
head(PCA.Factor.Rets)
```

##		Comp.1	Comp.2	Comp.3	Comp.4	
##	2000-05-31	0.37143392	-0.06946894	0.009199816	-0.124096094	0.0
##	2000-06-30	0.23212032	0.02312515	-0.076682683	0.060577666	-0.0
##	2000-07-31	0.05801505	0.04094138	0.031382432	0.001615707	-0.0
##	2000-08-31	0.06482303	0.08139716	-0.043479787	0.023226391	0.0
##	2000-09-29	0.41515651	0.02497782	-0.045495165	0.004944952	-0.0
##	2000-10-31	-0.41957345	-0.03783212	0.052618803	0.030764992	-0.0

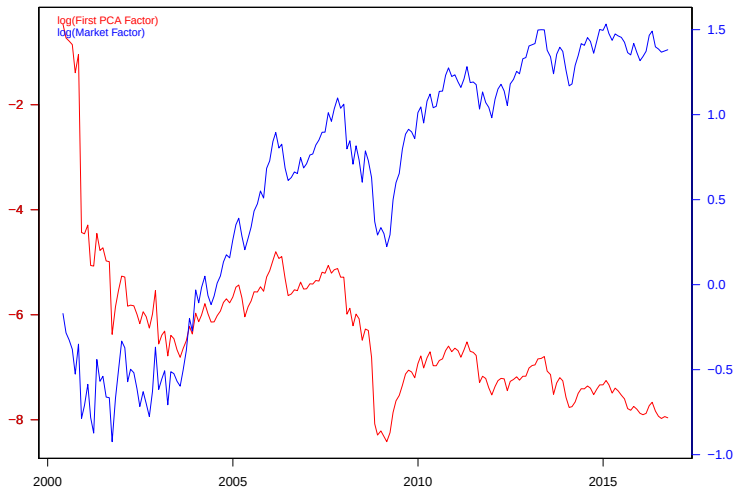
Statistical Factor Model

```
dat<-na.omit(merge(PCA.Factor.Rets,Bench.Rets))  
cor(dat)[6,1]
```

```
## [1] -0.955227
```

Statistical Factor Model

PCA Factor Model



Factor Investing for Turkish Equities

Empirical Findings

Factor Based Stock Selection Model
Betting Against Beta in BIST
Macro Factor Investing

The Secret To Successful Factor Investing

Long vs Short Term



PRESSMEDDELANDE
Press release

14 October 2013

The Prize in Economic Sciences 2013

The Royal Swedish Academy of Sciences has decided to award the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel for 2013 to

Eugene F. Fama
University of Chicago, IL, USA

Lars Peter Hansen
University of Chicago, IL, USA

Robert J. Shiller
Yale University, New Haven, CT, USA

"for their empirical analysis of asset prices"

Trendspotting in asset markets

There is no way to predict the price of stocks and bonds over the next few days or weeks. But it is quite possible to foresee the broad course of these prices over longer periods, such as the next three to five years. These findings, which might seem both surprising and contradictory, were made and analyzed by this year's Laureates, Eugene Fama, Lars Peter Hansen and Robert Shiller.

Beginning in the 1960s, Eugene Fama and several collaborators demonstrated that stock prices are extremely difficult to predict in the short run, and that new information is very quickly incorporated into prices. These findings not only had a profound impact on subsequent research but also changed market practice. The emergence of so-called index funds in stock markets all over the world is a prominent example.

If prices are nearly impossible to predict over days or weeks, then shouldn't they be even harder to predict over several years? The answer is no, as Robert Shiller discovered in the early 1980s. He found that stock prices fluctuate much more than corporate dividends, and that the ratio of prices to dividends tends to fall

Lars Peter Hansen developed a statistical method that is particularly well suited to testing rational theories of asset pricing. Using this method, Hansen and other researchers have found that modifications of these theories go a long way toward explaining asset prices.

Another approach focuses on departures from rational investor behavior. So-called behavioral finance takes into account institutional restrictions, such as borrowing limits, which prevent smart investors from trading against any mispricing in the market.

The Laureates have laid the foundation for the current understanding of asset prices. It relies in part on fluctuations in risk and risk attitudes, and in part on behavioral biases and market frictions.

Eugene F. Fama, U.S. citizen. Born 1939 in Boston, MA, USA. Ph.D. 1964 from University of Chicago, IL, USA. Robert R. McCormick Distinguished Service Professor of Finance at University of Chicago, IL, USA.
www.chicagobooth.edu/faculty/directory/f/fama-e-fama

Lars Peter Hansen, U.S. citizen. Born 1952 in Urbana, IL.

Simple But Not Easy



Source: Greenblatt (2012)

THE SECRET TO VALUE INVESTING

Database

Database

BIST largecap stocks:

Open/High/Low/Close prices

Various factor variables